Unit-2: Random Variables and Probability Distributions

The word "probability" is commonly used to denote "the chance of happening of an event".
For example, statements like,
I have a fair chance (i.e., a reasonable probability of success) of getting admission.
In tossing a coin there is an even chance that a head may come up.
In each case, we are not sure of the outcome, but we wish to judge (or estimate) the chances of our predictions coming true. The study of probability provides a mathematical framework for such assertions (declaration/positive statement).

Elementary definitions

- Experiment: A process/procedure we perform that generates some result. For example, tossing a coin, rolling a die, etc.
- Outcome: An outcome is a possible result of an experiment. Let us consider an example of tossing a coin two times. Here, four possible outcomes are (H, T), (T, H), (T, T), and (H, H).
- Sample space: The set of "all possible" distinct outcomes of an experiment is called the sample space and is denoted by S. For example, for tossing a coin, the sample space S={H,T}. Note that a sample space S can be either "discrete" or "continuous".
- When the sample space S is countable, such as when it includes all integers between 1 and 9, it is known as a discrete sample space.
- When the sample space S is uncountably infinite, such as when it includes all real numbers between 1 and 9, it is known as a continuous sample space.
- Event: An event is a subset of a sample space. For example, for tossing a coin 3 times, the sample space is S={HHH, HHT, HTH, THH, HTT, THT, TTH, TTT}. Let A be the event of getting exactly two heads. Then $A =$ {HHT, HTH, THH}.
- Mutually exclusive events: Two events are said to be "mutually exclusive" if both cannot occur at the same time. For example, in a single coin toss, either a head comes up or a tail comes up, but not both.
- Mathematically, if two events A and B are mutually exclusive, then $A \cap B = \emptyset$.
- Equally likely event: The events are said to be "equally likely" if the chance of happening is equal of all events. For example, If we toss a coin, there are equal chances of getting a head or a tail. Hence, getting a head or a tail by tossing a coin are equally likely events.

- Exhaustive events: Two events A and B are known as exhaustive events if the union of A and B gives the sample space, i.e., $A \cup B = S$.
- For example, if a die is rolled, then the event of getting a prime number and an odd number are exhaustive events or not?

  Sample space = {1,2,3,4,5,6}.
  Event of getting a prime number = {1,2,3,5}.
  Event of getting an odd number = {1,3,5}.
  Union of both events = {1,2,3,5}.
  Both events together do not equal to the sample space. Hence, these events are not exhaustive.

- To find the probability of equally likely events, the following formula is used.

  $P(E) = \frac{n(E)}{n(S)}$ Here, $n(E)$ is total number of favorableevents; and $n(S)$ is a total number of events in sample space.

  Note that,

1. Probability of a certain (sure) event is one.
2. Probability of an impossible event is zero.

The probability of non-occurrence of an event E (called its failure) denoted by

$P(\bar{E}) = \frac{n(S)-n(E)}{n(S)} = 1 - \frac{n(E)}{n(S)} = 1 - P(E)$. Therefore, $P(E) + P(\bar{E}) = 1$

Axioms (statements that are always accepted as true) of probability

Axiom 1: For any event E, $0 \leq P(E) \leq 1$

Axiom 2: Let S be a sample space. Then $P(S) = 1$

Axiom 3: If E_1 and E_2 are mutually exclusive events, then

$$P(E_1 \cup E_2) = P(E_1) + P(E_2).$$

More generally, if $E_1, E_2, \dots, E_n, \dots.$ mutually exclusive events, then

$$P(E_1 \cup E_2 \cup \dots) = P(E_1) + P(E_2) + \cdots$$

For example, consider,

Let A be the event of getting tails when we flip a fair coin.

Axiom 1: $P(A) = \frac{1}{2}$. Since $P(A)$ is between 0 and 1, Axiom 1 holds.

Axiom 2: S ={Head, Tail}. P(S)= $\frac{1}{2} + \frac{1}{2}$ = 1. So, Axiom 2 holds.

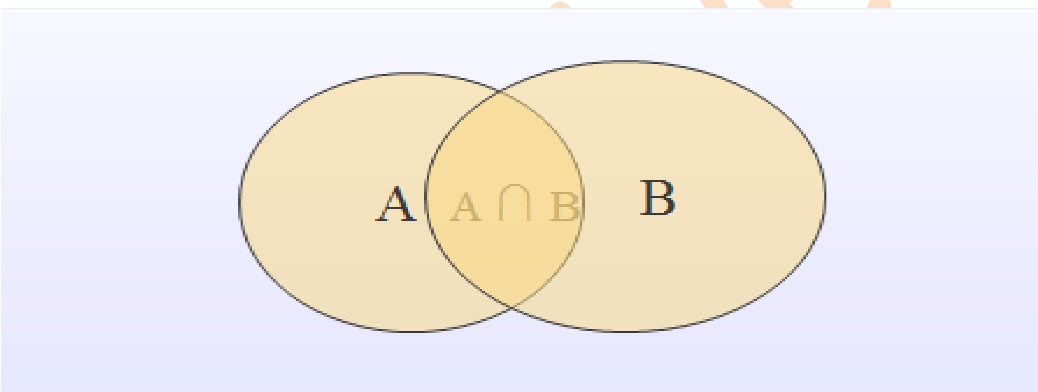
Axiom 3: Getting a head and a tail are two mutually exclusive events. Then

P({Head} U {Tail}) = P({Head})+P({Tail})= $\frac{1}{2} + \frac{1}{2}$= 1 . So, Axiom 3 holds.

Additive theorem for probability

Statement: If A and B are any two arbitrary events (not necessarily mutually exclusive) of a sample space S, then

$$P(A \cup B) = P(A) + P(B) - P(A \cap B).$$



Additive theorem for mutually exclusive events

Statement: If A and B are two mutually exclusive events, then the probability of either A or B is given by $P(A \cup B) = P(A) + P(B)$.

For example, consider,

1. A card is drawn from a pack of 52, what is the probability that it is a king or a queen?

Solution: Let Event A = Draw of a card of king. Event B = Draw of a card of queen.

P (card draw is king or queen) = P (card is king) + P (card is queen)

$$P(A \cup B) = P(A) + P(B) = \frac{4}{52} + \frac{4}{52} = \frac{2}{13}$$

Random variables

Let us take an example of tossing a coin.

The sample space is S ={heads, tails}

Let us give them the values: heads=0 and tails=1, and here we have a random variable X.

That is, $X = \begin{cases} 0, & if\, \omega \in S \text{ is heads} \\ 1, & if\, \omega \in S \text{ is tails} \end{cases}$

In short, $X = \{0,1\}$

- A Random variable is a set of **possible values** from a random experiment.
- A random variable X on a sample space S is a function $X: S \longrightarrow R$ from S to the set of real numbers R, which assigns a real number $X(S)$ to each sample point ω of sample space S .
- A random variable X is a function from a sample space S into the real numbers R.

Types of random variables

Random variables are classified into the following two categories:

- Discrete random variable
- Continuous random variable

Discrete random variable:

A random variable X is said to be discrete if it assumes a finite or countable number of distinct possible values. Example:

- The number of people going to a cricket match.
- Number of people born in January.
- Note that in all the examples mentioned above, we cannot have values like 1.1 or 1.2 or 2.1, and so on.

Continuous random variable:

A random variable X is said to be continuous if it assumes infinite number of distinct possible values.

Example:

- Average age of people born in January.
- Average weight of people born in January.

Functions of random variables

- A random variable is a rule that assigns a numeric value to every possible outcome in a sample space S.
- A function of a random variable is another random variable that is derived from the original variable by applying a mathematical function to it.
- These functions transform the original random variable into a new one with different properties, such as its mean, variance, and probability distribution.
- The main functions of random variables include: Probability Density Function (PDF); Cumulative Distribution Function (CDF); Expected Value; Variance; Standard Deviation; and Moment Generating Function.

Binomial Distribution:

- Binomial distribution is a discrete probability distribution.
- The binomial random variable X is the number of successes in Bernoulli trials with the number of times that the event has occurred out of n trials. The possible values of X are $0,\ 1,\ 2, \dots, n$.
- Since X take only integer values X is discrete.
- Thus, binomial distribution is a discrete probability distribution.
- For a random experiment E, if event A happens, then we call it a success otherwise it is a failure.
- We associate a probability of success $P(A) = p$, and the probability of failure defined as $P(\bar{A}) = q = 1 - p$.

Examples that give rise to binomial distribution:

- Throwing a dice.
- Tossing of coins.

A random variable X is said to follow the binomial distribution if the probability distribution function is given by $P(X=r)=P(r)=nC_r p^r q^{n-r},\ r=0,\ 1,\ 2,\dots,n$ where $q=1-p$.

Note:

1. $\sum_{r=0}^{n} P(X=r)=\sum_{r=0}^{n} P(r)=\sum_{r=0}^{n} nC_r p^r q^{n-r}=(p+q)^n=1.$
2. The frequency function of the binomial distribution is defined by $f(r)=N*P(r)$, where N is the number of times the experiment is repeated.

Mean and Variance of Binomial Distribution

Mean $=$ expectation $=\mu=E(x)$

$$=\sum_{x=0}^{n} xP(x)$$

$$=\sum_{x=0}^{n} x*nC_x p^x q^{n-x}$$

$$=\sum_{x=0}^{n} x\,\frac{n!}{x!(n-x)!}\,p^x q^{n-x}$$

$$=np\sum_{x=1}^{n}\ \frac{(n-1)!}{(x-1)!(n-x)!}\,p^{x-1}q^{(n-1)-(x-1)}$$

$$=np\sum_{x=1}^{n}\ (n-1)C_{(x-1)}\,p^{x-1}q^{(n-1)-(x-1)}$$

$$=np(p+q)^{n-1}=np(1)=np.$$

Variance $=\sigma^2\ =\sum_{x=0}^{n}(x-\mu)^2P(x)$

$$=\sum_{x=0}^{n}(x^2-2\mu x+\mu^2)P(x)$$

$$=\sum_{x=0}^{n} x^2\ P(x)-2\mu\sum_{x=0}^{n} x\ P(x)+\mu^2\sum_{x=0}^{n} P(x)$$

$$=\sum_{x=0}^{n} x^2\ P(x)-2\mu.np+\mu^2.1$$

$$=\sum_{x=0}^{n} x^2\ P(x)-(np)^2 \qquad \underline{\qquad\qquad}(1)$$

Consider $\sum_{x=0}^{n} x^2 \ P(x)$

$$= \sum_{x=0}^{n} x^2 . nC_x p^x q^{n-x}$$

$$= \sum_{x=0}^{n} x^2 \ \frac{n!}{x!(n-x)!} \ p^x q^{n-x}$$

$$= \sum_{x=1}^{n} \ [x(x-1)+x] \frac{n!}{x!(n-x)!} \ p^x q^{n-x}$$

$$= \sum_{x=1}^{n} \ \ x(x-1). \frac{n!}{x!(n-x)!} \ p^x q^{n-x} + \sum_{x=1}^{n} \ \ x \frac{n!}{x!(n-x)!} \ p^x q^{n-x}$$

$$= \sum_{x=2}^{n} \ \frac{n!}{(x-2)!(n-x)!} \ p^x q^{n-x} + np$$

$$= n(n-1)p^2 \sum_{x=2}^{n} \ \frac{(n-2)!}{(x-2)!((n-2)-(x-2))!} \ p^{x-2} q^{(n-2)-(x-2)} + np$$

$$= n(n-1)p^2 \sum_{x=2}^{n} \ .(n-2)C_{(x-2)} \ p^{x-2} q^{(n-2)-(x-2)} + np$$

$$= n(n-1)p^2 (p+q)^{n-2} + np$$

$$= n(n-1)p^2 + np, \text{ since } p+q=1$$

Thus variance $= \sigma^2 = n(n-1)p^2 + np - n^2p^2 = np - np^2 = np(1-p) = npq.$

Binomial Distribution – Example

1. Assume that 50% of all engineering students are good in mathematics. Determine the probabilities that among 18 engineering students

   (i) exactly 10,

   (ii) at least 10,

   (iii) at most 8,

   (iv) at least 2 and at most 9, are good in Maths.

Solution:

Let X = number of engineering students who are good in Maths.

p = probability of a student good in Maths = $50\% = \frac{50}{100} = \frac{1}{2}$

$n = 18$ and $q = 1 - p = \frac{1}{2}$

$$P(x) = nC_x p^x q^{n-x} = 18C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{18-x}$$

(i) Exactly 10 students are good in Maths out of 18: $P(x = 10) = 18C_{10} \left(\frac{1}{2}\right)^{10} \left(\frac{1}{2}\right)^{18-10} = 0.1670$

(ii) At least 10 students are good in Maths out of 18: $P(x \geq 10) = P(11) + P(12) + \cdots + P(18)$

$$= 18C_{11} \left(\tfrac{1}{2}\right)^{11} \left(\tfrac{1}{2}\right)^{7} + \cdots + 18C_{18} \left(\tfrac{1}{2}\right)^{18} \left(\tfrac{1}{2}\right)^{0} = 0.4073$$

(iii) At most 8 students are good in Maths out of 18: $P(x \leq 8) = P(1) + P(1) + \cdots + P(1)$

$$= 18C_1 \left(\tfrac{1}{2}\right)^{1} \left(\tfrac{1}{2}\right)^{17} + \cdots + 18C_8 \left(\tfrac{1}{2}\right)^{8} \left(\tfrac{1}{2}\right)^{10} = 0.4073$$

(iv) At least 2 and at most 9 are good in Maths out of 18: $P(2 \leq x \leq 9) = P(2) + P(3) + \cdots + P(9)$

$$= 18C_2 \left(\tfrac{1}{2}\right)^{2} \left(\tfrac{1}{2}\right)^{16} + \cdots + 18C_9 \left(\tfrac{1}{2}\right)^{9} \left(\tfrac{1}{2}\right)^{9} = 0.5920$$

Poisson Distribution

Poisson distribution is the discrete probability distribution of a discrete random variable x, which has no upper bound.

It is defined for non-negative values of x as follows:

$$f(x, \lambda) = P(X = x) = \frac{\lambda^x e^{-\lambda}}{x!}, \text{ for } x = 0,\ 1,\ 2,\ \ldots$$

Here $\lambda > 0$ is called the parameter of the distribution.

Examples that give rise to Poisson distribution:

- Number of printing mistakes per page
- Number of accidents on a highway
- Number of defectives in a production centre
- Number of telephone calls during a particular time

Mean and Variance of Poisson Distribution

Mean: $E(X) = \sum_{x=0}^{\infty} xP(x)$

$= \sum_{x=0}^{\infty} x.\frac{\lambda^x e^{-\lambda}}{x!}$

$= \lambda e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^{x-1}}{(x-1)!}$

$= \lambda e^{-\lambda}\left(1 + \frac{\lambda}{1!} + \frac{\lambda^2}{2!} + \cdots\right)$

$= \lambda e^{-\lambda} e^{\lambda} = \lambda$

Variance: $E(X^2) = \sum_{x=0}^{\infty} x^2 P(x) - \lambda^2$

Consider $\sum_{x=0}^{\infty} x^2 P(x) = \sum_{x=0}^{\infty} x^2.\frac{\lambda^x e^{-\lambda}}{x!} = \sum_{x=0}^{\infty}[x(x-1)+x].\frac{\lambda^x e^{-\lambda}}{x!}$

$= e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x.x(x-1)}{x!} + e^{-\lambda} \sum_{x=0}^{\infty} \frac{\lambda^x.x}{x!}$

$= e^{-\lambda} \sum_{x=2}^{\infty} \frac{\lambda^x}{(x-2)!} + e^{-\lambda} \sum_{x=1}^{\infty} \frac{\lambda^x}{(x-1)!}$

$= \lambda^2 e^{-\lambda}\left(1 + \frac{\lambda}{1!} + \frac{\lambda^2}{2!} + \cdots\right) + \lambda\, e^{-\lambda}\left(1 + \frac{\lambda}{1!} + \frac{\lambda^2}{2!} + \cdots\right)$

$= \lambda^2 + \lambda$

Thus variance $E(X^2) = \lambda^2 + \lambda - \lambda^2 = \lambda$

Poisson Distribution – Example

1. Suppose that a book of 600 pages contains 40 printing mistakes. Assume that these errors are randomly distributed throughout the book and the number of errors per page has a Poisson distribution. What is the probability that 10 pages selected at random will be free of errors?

Solution: $p = \frac{40}{600} = \frac{1}{15}, \quad n = 10, \quad \lambda = np = 10 \times \frac{1}{15} = \frac{2}{3}$

$$P(x) = \frac{\lambda^x e^{-\lambda}}{x!} = \frac{\left(\frac{2}{3}\right)^x e^{-\frac{2}{3}}}{x!}$$

$$P(x=0) = \frac{\left(\frac{2}{3}\right)^0 e^{-\frac{2}{3}}}{0!} = e^{-\frac{2}{3}} = 0.51$$